

HW 4

1

Derive a set of basic cells to implement the iterative logic array in CMOS.

✓ Answer ✓

We would need to implement:

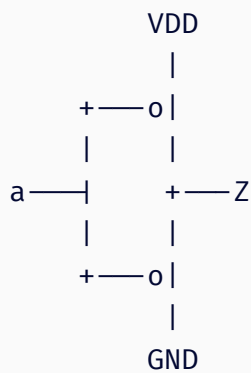
1. Inverter
2. 4-input ANDOR22'
3. 2-input NAND

2

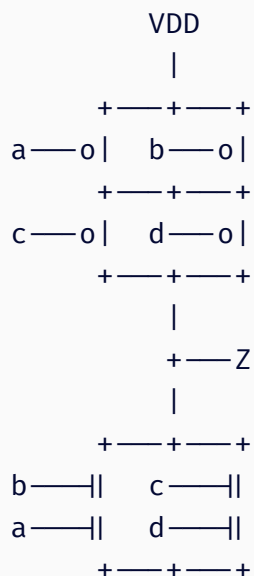
For each of the basic cells, design a static CMOS transistor level network?

✓ Answer

Inverter

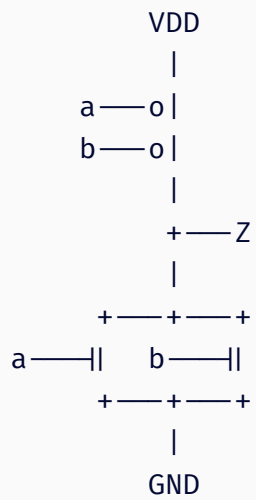


and-or22'

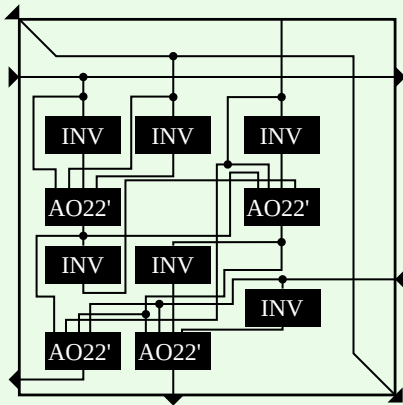


GND

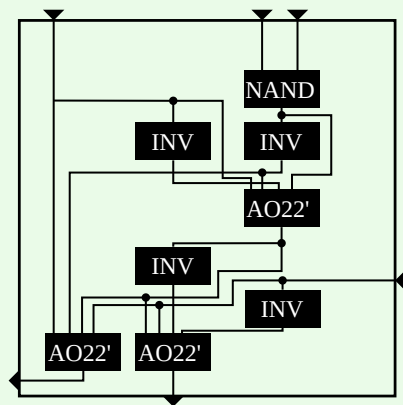
NAND



CAS



RC

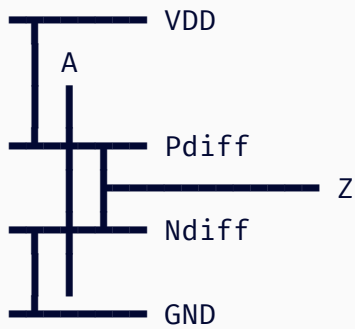


3

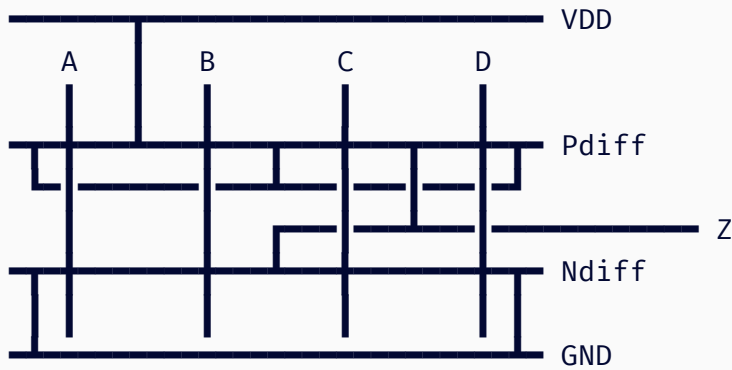
Derive a stick diagram for each cell.

✓ Answer

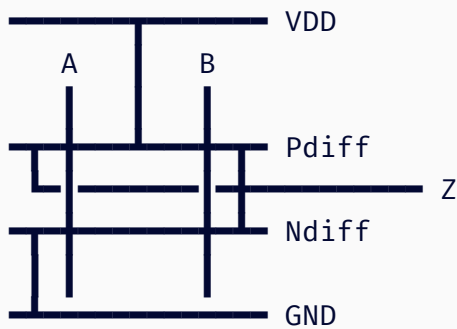
Inverter for Nichols



and-or22'



NAND



4

For your divider, identify the path the exhibit the worst case delay? (Explain)

✓ Answer

Likely from M_3 to R_3 as it has to make its way rightwards then leftwards through each CAS on each row before propagating downwards through the Q_i outputs. Once at the bottom row, it passes through one remainder correction before obtaining R_3 . Since the signal first passes right before propagating left on each row of CAS's, the leftmost CAS will have the most delay out of the row.